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Objectives: Neck injuries in football are attributed to helmet-to-helmet contact with youth players being at greatest risk. In 2014 the National Federation of State High School Associations (NFHS) implemented rules defining illegal contact against a defenseless player above the shoulders to reduce head and neck injuries in football players. This study evaluates whether rule implementation decreased rates of high school football neck injuries presenting to the emergency department (ED) pre-rule implementation (2009-2013) to post-rule implementation (2015-2019).

Methods: Data were queried from the National Electronic Injury Surveillance System for high school football players 14 to 18 years old diagnosed with a neck injury from January 1, 2009 to December 31, 2019. Narratives in the data were reviewed for mechanism of injury, setting, loss of consciousness (LOC), and type of injury.

Results: Between 2009 and 2019, an estimated 47,577 high school football neck injuries were diagnosed in EDs across the United States. 52.0% of neck injuries were sustained during competition compared to 48.0% during practice. A statistically significant (P=0.004) decrease in neck injuries was realized from pre-rule implementation to post-rule implementation with averages of 5,278 and 3,481 injuries per year, respectively. Helmet-to-helmet neck injuries significantly (P=0.04) decreased from pre- to post-rule implementation with averages of 851 and 508 injuries per year, respectively. Neck injuries sustained via other mechanisms were not affected by the 2014 rule implementation.

Conclusion: This study is the first to identify a decrease in overall and helmet-to-helmet related neck injuries diagnosed in the ED following the 2014 NFHS targeting rule implementation.

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These findings add to the growing literature regarding the importance and efficacy of rule implementation in reducing sports-related neck injuries.

Keywords: Youth, spine, NEISS, cervical, injury prevention, athletics, pediatrics

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Introduction

Fifteen percent of all cervical spine injuries occur during sports, which is the fourth highest cause behind motor vehicle accidents, violence, and falls[1, 2]. Football is an inherently physical sport, putting athletes at greater susceptibility of injury[3–5]. As a result, football players exhibit the highest rate of head, neck and cervical spine injuries compared with athletes of other sports across all age levels[6, 7]. In addition, football constitutes the greatest number of cervical sprains of all sports presenting to the emergency department (ED), suggesting a greater severity of injury[8].

With 90.8% of neck injuries in NCAA football resulting from direct contact, reducing helmet-to-helmet contact and axial compression is essential in minimizing neck and cervical spine injuries[6]. Neck injuries in football have been attributed to helmet-to-helmet contact when blocking, tackling, or head butting[5]. This often occurs when the neck flexes forward prior to contact, thus resulting in a straightened cervical spine. This neck positioning places the player at increased risk of injury from an axially directed force via the lack of shock reduction provided by the natural cervical lordotic curvature[9]. Axial loading from helmet-to-helmet contact is the greatest risk factor for cervical spine injury and quadriplegia compared to hyperextension or hyperflexion[5, 9, 10]. As a result, 96.3% of spinal cord injuries in football players occur at the cervical level[11]. Other common mechanisms include forced full range extension, compression of the brachial plexus or Erb's point (stingers), direct trauma, or repeated impacts.

Compared to their adult counterparts, younger athletes have reduced neck strength and musculature as head development occurs at a faster rate than the body[12]. This imbalance

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puts youth athletes at greater risk for injuries involving the neck. The decreased neck strength makes youth athletes prone to forced end-range neck positions and soft tissue damage[13–15]. Additionally, youth athletes may be more susceptible to head and neck injuries due to their reduced capacity to control abrupt head deceleration and continued torso momentum upon impact[16].

Rule implementation to reduce neck injuries in sports has gained momentum in recent years. The International Ice Hockey Federation set rules prohibiting checking and charging from behind, resulting in decreased spinal cord and overall injuries[17, 18]. In 2014, the National Federation of State High School Associations (NFHS) implemented Rule 2-43, the “targeting rule,” to bring awareness to the action of targeting and minimizing head injuries in football players. The “targeting rule” is defined as “an act of taking aim and initiating contact with an opponent above the shoulders with the helmet, forearm, hand, fist, elbow or shoulders." The “defenseless player” (Rule 2-32-16) was defined as “a player who, because of his physical position and focus of concentration, is especially vulnerable to injury,” to expand the use of the targeting rule. Although implemented to minimize concussions, these rules may also indirectly reduce neck injuries. To our knowledge, trends in ED diagnosed neck injuries of high school football players pre- and post-rule implementation have not been described.

Recent studies have highlighted the impact of rule change on rates of injury in athletics using the National Electronic Injury Surveillance System (NEISS) database[18]. However, despite the recent focus on reducing helmet-to-helmet contact, there is a paucity of research on the effect these rules have on rates of neck and cervical spine injury. One study identified a decreasing rate of head and neck injuries in youth football players from 2008-2017 using the

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NEISS database, which the authors attributed to improvements in equipment quality and rule implementation[19]. However, narratives of each injury were not reviewed to exclude data that did not pertain to organized football, thus resulting in less reliable heterogeneous results with significant confounding.

The purpose of this study was to identify whether the 2014 NFHS rule implementation decreased rates of high school football neck injuries presenting to the ED by comparing trends from pre-rule implementation (2009-2013) to post-rule implementation (2015-2019). We hypothesized that rates of overall and helmet-to-helmet neck injuries were lower following the 2014 rule implementation.

Materials and Methods

Database Background

Injury data were queried from the NEISS, a national public database maintained by the U.S. Consumer Product Safety Commission (CPSC). The NEISS collects information from EDs of 100 hospitals with a minimum of six beds and 24-hour services that are selected as a representative probability sample of the 5,000 U.S. hospital EDs. National estimates (NE) are calculated based on the assigned statistical sample weight of each hospital to generate reliable epidemiological data.

All football-related injuries (product code 1211) from January 1, 2009, to December 31, 2019 were queried for patients 14 to 18 years of age who sustained a neck injury (product code 89). Initially, 1,487 cases (NE = 50,878) were queried.

Data Collection

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Patient age, patient sex, and date of injury were provided in the queried data. The narrative associated with each injury was reviewed to identify the mechanism of injury, type of injury, setting of injury, loss of consciousness (LOC), and whether there was concomitant fracture. The authors defined and categorized mechanism of injury as helmet-to-helmet, helmet-to-ground, helmet-to-body part, pulling of the facemask, other, or not specified. Setting of injury was either practice, competition, or not specified. Type of injury was categorized as contusions, disc injury, nerve injury, sprains, strains, vertebral fractures, other, or not specified.

Exclusion Criteria

The narrative of each injury was reviewed to identify and exclude injuries sustained outside of organized high school tackle football. Injuries sustained while not wearing full equipment were excluded. Injuries that mentioned locations other than a football field were excluded (e.g., home, street, park). Injuries that took place in a setting other than organized competition or practice were excluded (e.g., physical education/gym class, pick-up games). Injuries that occurred during organized high school tackle football but were not related to participation (e.g., getting hit by errantly thrown ball on the sideline, injury while celebrating) were excluded.

Statistical Analysis

Statistical analyses were performed using STATA/MP Software 13.0 (StataCorp, College Station, TX) and SPSS (IBM SPSS Statistics for Windows Version 26.0. Armonk, NY:IBM Corp). Unpaired Student's T-tests and Poisson regressions were used to compare trends pre- and post-rule implementation. Statistical significance was set at $P<0.05$. The year 2014 was removed from analysis to control for rule-change bias when comparing trends pre- and post-rule

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implementation. NE were determined by multiplying the respective statistical weight of each hospital by the raw, queried data.

Results

Between January 1, 2009 and December 31, 2019 there were 1,487 cases (NE = 50,878) of ED diagnosed neck injuries related to football. After sorting through the narratives using inclusion and exclusion criteria, 1,388 (NE = 47,577) were included for analysis (Figure 1). Mean age at diagnosis was 15.5 years (range: 14-18 years). 98.8% (NE = 46,989) of the patients were males, and 1.2% (NE = 588) were females (Table 1). For narratives that specified setting, 52.0% of neck injuries occurred during competition compared to 48.0% during practice.

Review of the mechanism of injury through the narrative provided national estimates for not specified (NE = 31,885; 67.0%), helmet-to-helmet (NE = 7,057; 14.8%), other mechanism (NE = 3,722; 7.8%), helmet-to-ground (NE = 2,530; 5.3%), helmet-to-body part (NE = 1,497; 3.1%), and pulling the facemask (NE = 885; 1.9%) (Table 2). The particular body parts involved in helmet-to-body concussive episodes included the knee (NE = 363; 24.2%), shoulder (NE = 361; 24.1%), elbow (NE = 264; 17.6%), chest (NE = 227; 15.2%), foot (NE = 120; 8.0%), abdomen (NE = 95; 6.3%), and leg (NE = 67; 4.5%).

Type of neck injuries included strains (NE = 28,464; 59.8%), not specified (NE = 8,496; 17.9%), sprains (NE = 7,000; 14.7%), contusions (NE = 1,952; 4.1%), vertebral fractures (NE = 960; 2.0%), nerve injuries (NE = 423; 0.9%), other (NE = 247; 0.5%), and disk injuries (NE = 35; 0.1%).

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A statistically significant (*P*=0.004) decrease in neck injuries was realized from pre-rule implementation to post-rule implementation with yearly averages of 5,278 and 3,481 neck injuries, respectively (Table 3). Helmet-to-helmet neck injuries significantly (*P*=0.04) decreased from pre- to post-rule implementation with yearly averages of 851 and 508 neck injuries, respectively. Not specified mechanisms of neck injury demonstrated a statistically significant (*P*=0.002) decrease in yearly averages from 3,546 to 2,265 neck injuries, respectively. Neck injuries sustained via other mechanisms were not affected by the 2014 rule implementation

From the decade of 2009 to 2019, there was a statistically significant (*P*=0.0003) decrease in neck injuries (coefficient = -0.0121; 95% CI: [-0.0187 to -0.0055]). When utilizing the pre-rule implementation period as the reference, a statistically significant (*P*<0.0001) decrease in neck injuries was realized following the 2014 rule implementation (coefficient = -0.3454; 95% CI: [-0.3895 to -0.3013]).

Discussion

Although concussions have garnered the majority of attention in recent years, the severity and impact of neck injuries cannot be overlooked. This study is the first to identify a significantly decreased yearly average number of overall (5,287 vs. 3,481) and helmet-to-helmet (851 vs. 508) neck injuries in high school football players following the 2014 NFHS rule implementation. This study also identifies a statistically significant decrease in neck injuries following the 2014 rule implementation after taking into account the pre-existing decline. Importantly, the decrease in neck injuries following the 2014 rule implementation is

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proportionally larger than the progressive decline in reported high school football participation in the US[20].

Rates of ED neck injuries in this study differ from those previously reported. The consistently high incidence of neck injuries from 2009-2013 contradict that of Scheffler et al., who found a decreasing trend in head and neck injuries beginning in 2008[19]. However, Scheffler et al. analyzed ED diagnoses in all youth athletes less than 18 years of age and did not differentiate between head and neck injuries. Subsequently, this discrepancy may be attributed to the differences in patient demographics and injury diagnosis influencing the data.

This study also reports the distribution of different types of neck injuries sustained in high school football players presenting to the ED. Neck strains constituted the majority of overall injuries in this study (59.8%). Radiographic imaging may be needed to rule out instability from severe ligamentous disruption in neck strains, which is difficult to evaluate in youth athletes[9]. As a result, athletes exhibiting symptoms of acute, severe neck strain may be removed from play and admitted to the ED for subsequent evaluation.

Other severe injuries, such as fracture (2.0%), nerve injury (0.9%), and disk injury (0.1%), occurred at much lower rates compared to strains. These findings differ from rates seen in NCAA football players, in which 64.5% of neck injuries were brachial plexus-related, while 28.7% were a combination of strains, sprains, or contusions[6]. However, similar rates of disc injury and fractures occurred in NCAA football players (1.8% and 1.5%, respectively) as in this study. The higher rate of nerve-related injuries in Deckey et al. may be attributed to the high incidence of college football players experiencing brachial plexus injuries (stingers)[21, 22]. These injuries are often transient with 63.8% resolving in under 24 hours, thus allowing for

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sideline treatment by an athletic trainer without the need for ED evaluation[9, 23]. Subsequently, the low proportion of nerve injuries in our study may represent a selection bias as cases of greater severity and prolonged neurologic findings resulting in presentation to the ED are much less common in youth football.

Interestingly, there was a large increase in overall and helmet-to-helmet neck injuries in 2019 after a decrease from 2015-2018. Although it is unclear why this occurred, our hypothesis is that awareness of neck and spinal cord injuries increased. As a result, athletes who experienced a minor injury may have been more likely to report to the ED for further evaluation, constituting the large proportion of strains in this study.

A similar proportion of neck injuries occurred in practice (48.0%) as in competition (52.0%) in this study. This finding is comparable to Deckey et al, who identified a 50.5% of neck injuries occurring in practice and 49.5% occurring in competition for NCAA football players. Interestingly, this distribution is different than that of concussions despite a similar mechanism impact and injury. Previous research has identified concussions occurring at a much higher rate in competition than practice[24, 25]. The similar incidence of neck injuries in both practice and games highlights the importance of minimizing neck injuries in both settings of play.

This study is not without limitations. The inherent nature of using a national database limited data collection to what was provided in the query. Narratives, which are dependent on providers recording sufficient information, provide in-depth information on mechanism of injury not provided by codes. Due to the lack of information, we were unable to categorize injuries as occurring on offense, defense, or special teams. Additionally, a large proportion of the injuries in our study (67.0%) did not contain adequate information to determine mechanism

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and were thus labeled as “not specified.” We included these data, rather than eliminate, to provide better insight into the overall trends of neck injury during the study period. Due to the nature of the NEISS, this study is limited to athletes presenting to the ED and does not include players that were diagnosed and treated by athletic trainers, primary care physicians, specialists, or urgent care clinics. Subsequently, this study may reflect injuries of greater severity. This may explain the discrepancy between the greater number of brachial plexus injuries in Deckey et al. compared to nerve injuries in this study[6]. Lastly, these data represent ED diagnoses stratified across 100 US hospitals and are not representative of the true incidence of neck injuries. However, the national estimates provide reliable analyses of trends in high school neck injuries.

Conclusion

This study is the first to identify a significant decrease in annual incidence of overall and helmet-to-helmet associated high school football neck injuries diagnosed in the ED following the 2014 NFHS rule implementation. Although these findings highlight the importance and efficacy of rule implementation on reducing neck injuries, further research is warranted to identify causation.

Disclosure of Interest

The authors report no conflicts of interest

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Figure Legends

Figure 1: Decreasing Trend in National Estimates of Neck Injury Totals by Year

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Tables

Table 1: Study Characteristics and Demographic Data

Patient Characteristics	National Estimate	Raw
Neck Injuries	47,577	1,388
Loss of Consciousness	165 (0.3%)	4 (0.3%)
Mean Age (years)	15.52 (14-18)	15.45 (14-18)
Males	46,989 (98.8%)	1,369 (98.6%)
Females	588 (1.2%)	19 (1.4%)

Table 2: Mechanism Neck Injury Data

Mechanism	Neck Injury National Estimate (%)	Neck Injury Raw (%)
Not Specified	31,885 (67.0)	894 (64.4)
Helmet-to-Helmet	7,057 (14.8)	230 (16.6)
Other	3,772 (7.8)	111 (8.0)
Helmet-to-Ground	2,530 (5.3)	71 (5.1)
Helmet-to-Body Part	1,497 (3.1)	49 (3.5)
Pulling of Facemask	885 (1.9)	33 (2.4)
Total	47,577	1,388

Table 3: Average Neck Injuries Per Year Pre- vs. Post-Rule Implementation

Mechanism	Pre-Rule Implementation (2009-2013)	Post-Rule Implementation (2015-2019)	P-value
Overall	5,278	3,481	0.004
Helmet-to-Helmet	851	508	0.04
Not Specified	3,546	2,265	0.002

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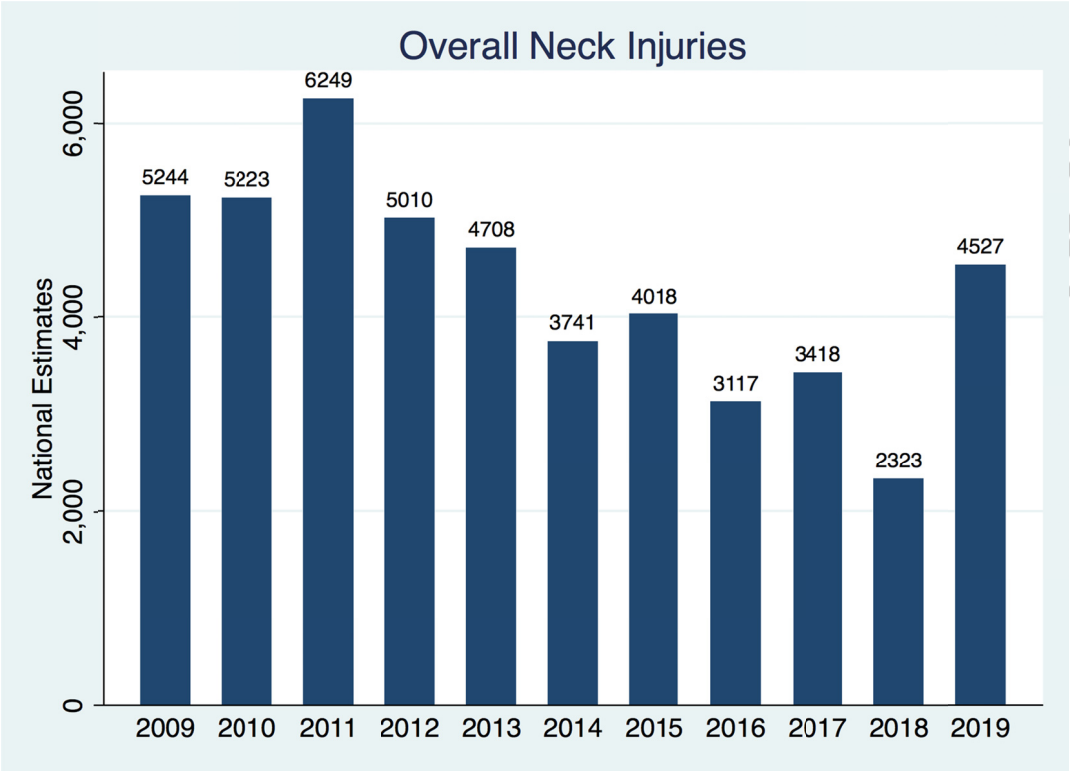
Figures

Figure 1:



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